

Smart Facility & Sustainability Manager

KOHLER-MITWPU AI Research Lab Program — Track 2 Case Study

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Problem & Approach

The Challenge

Commercial facilities lose significant water to undetected leaks. Manual inspection is too slow and inconsistent.

Our Solution

- Continuous fixture-level monitoring
Across the entire facility, around the clock
- Adaptive detection
Tuned per fixture, per hour — not one fixed threshold
- Transparent evidence
Behind every flagged issue
- Business-impact quantification
Tied directly to KOHLER's sustainability mission

System Architecture

Sensor Sim

Detection

SQLite DB

FastAPI

Dashboard

Tickets

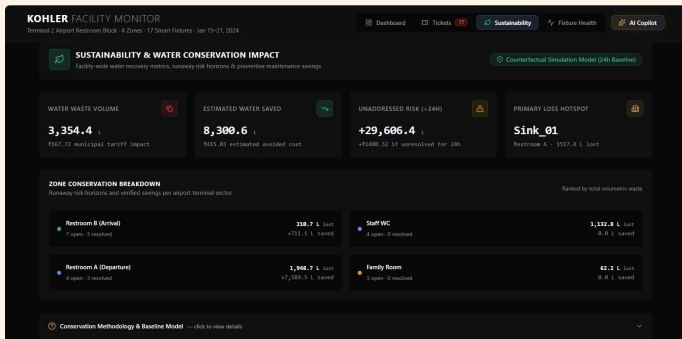
Sustainability

Fixture Health

AI Copilot

Detection Logic

FOUR-STEP DIFFERENTIATOR



Adaptive Baseline

Per-fixture, per-hour comparison against historical patterns

Multi-Signal Correlation

Occupancy + duration + sensor health cross-check

Slow-Drip Detection

Catches gradual leaks via rate-of-change analysis

Weighted Severity Scoring

Critical / High / Medium / Low — not binary

Explainability & Auditability

Plain-English Reasoning

Every ticket includes a clear explanation of why it was flagged

Evidence Strength Rating

Strong / Moderate / Weak — fully transparent, no guessing

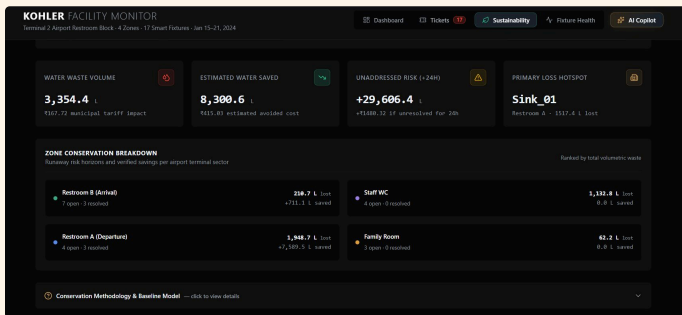
Real Workflow

Open → Dispatched → Resolved with resolution notes at every stage

No Black Box

Every decision is auditable and traceable

Sustainability & Business Impact



Automatic Impact Quantification

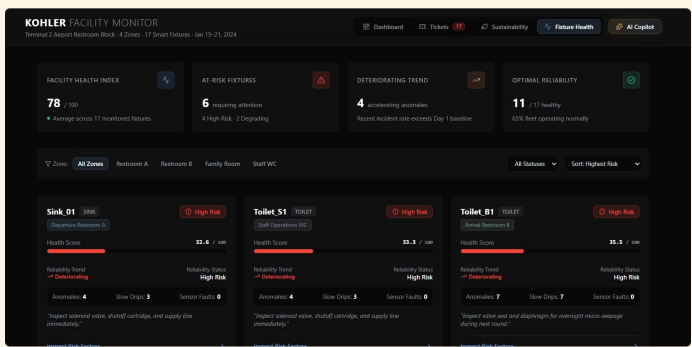
The system calculates water waste volume, estimated savings from prompt resolution, utility cost impact, and unaddressed risk — per zone and per fixture.

Every detection directly supports KOHLER's water conservation mission.

Additional Features

Fixture Health

Per-fixture risk score with trend direction and maintenance recommendations.



AI Copilot

Conversational interface grounded in live system data.

Ask about facility health in natural language

Query leak patterns and detection history

Explore sustainability impact on demand

Tech Stack & Deployment



Backend

FastAPI



Frontend

Next.js / React



Database

SQLite



AI

Google Gemini API

This architecture scales seamlessly to a full building portfolio — same detection logic, same transparency, same impact.

Thank you — Preksha